

Operations Manual 2026

Version 3.0, effective 30 July 2026. Owner: Dr. Sameer Deshpande, Medical Director, and Anjali Thomas, Chief Nursing Officer. Approved by Ritu Malhotra, Chief Executive Officer. This manual describes how Nephrova runs dialysis at all three centres. Where a detailed SOP or policy exists, this manual names it; the SOP or policy is the controlling document.

Chapter 1. About Nephrova

Nephrova Renal Care Pvt. Ltd. was founded in 2018. Our only service is outpatient haemodialysis for people with chronic kidney disease. We do not run wards, theatres or emergency departments. Every process in this manual exists to make each dialysis session safe, on time and well recorded.

We run 30 dialysis machines across three centres in Pune and treat about 238 patients who come two or three times a week. Most patients are funded by the PM-JAY government scheme; the others pay themselves or use private insurance.

Head office: Office 402, Sai Plaza, Baner Road, Pune 411045.

Chapter 2. Centres

Centre	Area	Dialysis machines	Active patients	Centre Manager
Nephrova Kothrud	Kothrud, Pune	12	96	Amit Gokhale
Nephrova Hadapsar	Hadapsar, Pune	10	80	Farah Khan
Nephrova Pimpri	Pimpri, Pune	8	62	Suresh Nair

Every centre runs two shifts, 06:30 to 14:30 and 14:00 to 22:00, from Monday to Saturday. Sunday is kept for maintenance, water plant disinfection and make-up sessions. Each centre has its own reverse osmosis water plant, a 125 kVA diesel generator and a UPS that carries every dialysis machine through a power cut until the generator takes over.

Patients are assigned to one home centre. Sessions move to another centre only under the Patient Transfer Protocol.

Chapter 3. The patient journey

3.1 Referral and admission

Patients are referred by their nephrologist. Before the first session we need the referral letter, the last three months of blood results, hepatitis B, hepatitis C and HIV screening, and payer details (PM-JAY card, insurance policy, or self-pay consent).

Patients positive for hepatitis B are dialysed only on the dedicated isolation machine at their centre.

3.2 Scheduling

Most patients dialyse three times a week for four hours. Twice-weekly schedules need the Medical Director's approval. Missed sessions are reported to the nephrologist the same day.

3.3 A dialysis session, step by step

1. The technician runs the machine's self test and checks the dialysate conductivity reading.
2. The nurse records weight, blood pressure, pulse and temperature, and agrees the fluid removal target.
3. Vascular access is connected: fistula needling, or a catheter connection under the Catheter Care

Bundle.

4. Blood pressure is recorded every 30 minutes during treatment.

5. At the end, the line is returned, the access is secured, weight is recorded and the treatment chart is signed.

3.4 Complications during treatment

Hypotension during dialysis is treated with normal saline given as a bolus into the blood line, and fluid removal is slowed.

Muscle cramps are managed by slowing fluid removal and stretching; the nurse records every episode.

Hyperkalaemia found on monthly bloods is treated with calcium gluconate in an emergency and reviewed by the nephrologist.

Fever or chills at the start of a catheter session is reported to the Infection Control Nurse the same day and blood cultures are taken.

Chapter 4. Dialysis machines

All machines are the Renalis R5 model. Biomedical engineering is led by Nikhil Joshi with technician Rahul Pawar.

4.1 Preventive maintenance

Centre	First half	Second half
Nephrova Kothrud	February	August
Nephrova Hadapsar	March	September
Nephrova Pimpri	April	October

Preventive maintenance covers the hydraulic circuit, blood pump, sensors, safety checks and electrical safety, and takes two days per centre.

4.2 Responding to alarms

Alarm	First response	If it repeats
Conductivity alarm	Pause treatment, check the concentrate and connections	After the second unexplained alarm, take the machine out of service and call biomedical engineering
Venous pressure alarm	Check the needle position and the venous line for kinks	Re-needle with the nurse in charge
Air detector alarm	Clamp lines, check the blood line for air	Re-prime the blood line
Blood leak alarm	Stop the blood pump and check the dialyser	Replace the dialyser and blood line

Every alarm is written in the centre's alarm log with the time, the machine and the action taken.

Chapter 5. Dialysis water

Dialysis water is made by a double-pass reverse osmosis plant at each centre and checked against ISO 23500 limits. SOP-WT-002 is the controlling procedure.

Parameter	Action level	Maximum
Bacteria	50 CFU/mL	100 CFU/mL

Endotoxin	0.125 EU/mL	0.25 EU/mL
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Each water plant and its loop receive chemical disinfection every month. A sample from the end of the loop is sent to the external laboratory every month.

A result above the maximum means: inform the Medical Director the same day, reduce or stop dialysis at that centre, move sessions under the Patient Transfer Protocol, disinfect, resample, and resume only after a result below the action level.

Chapter 6. Power

A UPS at each centre carries all dialysis machines through a mains failure. The diesel generator must start within 10 seconds. Generators run a monthly test on load; starter batteries are checked at the same time.

Chapter 7. Infection prevention

The Infection Control Policy, the Hand Hygiene Policy and the Catheter Care Bundle apply at every station. Hand hygiene and catheter care are audited monthly at every centre with a target of 90 percent.

A catheter-related bloodstream infection is treated with vancomycin after each session as prescribed by the nephrologist, and the case is reported to the Infection Control Nurse and the Quality Committee.

The infection rate is reported every month per 1,000 catheter days for each centre.

Chapter 8. Bio-medical waste

Waste is segregated at the station into yellow, red, white and blue containers as described in the Bio-Medical Waste Management SOP, and handed to the authorised treatment facility every 24 hours. The authorisation is valid until 31 March 2028.

Chapter 9. Medicines used in the unit

Medicine	Used for
Heparin	Preventing clotting in the circuit during dialysis
Erythropoietin	Anaemia of dialysis patients, given after the session
Iron sucrose	Iron deficiency, into the blood line
Normal saline	Hypotension during dialysis
Vancomycin	Catheter-related bloodstream infection, on prescription
Calcium gluconate	Hyperkalaemia emergencies

Anaemia in dialysis patients is treated with erythropoietin, with iron sucrose when iron stores are low.

Chapter 10. Quality and NABH

All three centres are accredited by NABH. The Quality Committee meets every month under the Medical Director and reviews water results, infections, alarms, complications, audits and incidents. Kiran Bhosale is the NABH coordinator.

Quality indicators reported monthly: treatment adequacy (Kt/V), missed sessions, hypotension episodes per 100 sessions, infections per 1,000 catheter days, water results, and hand hygiene compliance.

Chapter 11. Incidents and escalation

Every incident, near miss or equipment fault is reported the same day on an incident report. The Centre Manager informs the Medical Director at once if a patient was or could have been harmed, if dialysis must be reduced, or if water results are above limits.

Each incident with a systemic cause gets a root cause analysis with named owners and due dates, tracked by the Quality Committee.

Chapter 12. Who to contact

Role	Name
Chief Executive Officer	Ritu Malhotra
Medical Director (Nephrologist)	Dr. Sameer Deshpande
Chief Nursing Officer	Anjali Thomas
Biomedical Engineering Lead	Nikhil Joshi
Biomedical Technician	Rahul Pawar
Infection Control Nurse	Pooja Shinde
Quality Manager (NABH Coordinator)	Kiran Bhosale
Finance Head	Vivek Jain
HR Manager	Snehal Patil
Centre Manager, Kothrud	Amit Gokhale
Centre Manager, Hadapsar	Farah Khan
Centre Manager, Pimpri	Suresh Nair
Senior Dialysis Technician, Hadapsar	Ganesh More

Chapter 13. Controlling documents

SOP-WT-002 Dialysis Water Treatment; Patient Transfer Protocol; Infection Control Policy; Hand Hygiene Policy; Catheter Care Bundle; Bio-Medical Waste Management SOP; Code of Conduct; and the HR and finance policies issued from August 2026.